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A Contribution to the History of Modern Patent Legislation in the United States and in France

By DR. ING. FREDRIK NEUMEYER, STOCKHOLM

I

Introduction

The history of patent legislation is the ugly duckling of history. The political historian cannot see anything worth mentioning in this connection; the economic historian will probably confine his attention to the English Statute of Monopolies though he may discuss patents in reference to the industrial revolution of the 18th century; the legal historian regards it as a speciality hardly worthy of the time and space normally devoted to the extensive and traditional subjects of the law. And, finally, the history of engineering does not yet belong to an acknowledged branch of science in which research can be carried out on any significant scale.

This omission is in direct contrast to the importance which this field of law has in the history of nations. As patents serve the progress of engineering, each industrial country has sooner or later to encounter the principles of patent law. Throughout the Western hemisphere, parliaments have discussed the subject; and before the growth of democratic government, emperors, kings or the doges of the Italian Republics granted various sorts of privileges for inventions. Even churchmen such as Martin Luther have been forced to meditate upon such exclusive rights when disputing about the ethical aspects of monopoly. The attitude of the courts towards the legal protection of technical inventions also serves to illustrate the outlook which an epoch has on the rights of the individual, on the belief in industrial progress in general, and on the principles of the exchange of ideas and goods between different countries. The treatment of inventions and inventors by the legislators is also a symbol of the general economic policy of a country. What follows is an attempt to give some small proof of this assertion.

II

United States of America

The protracted and intense conflicts, both political and military, which led to the birth of the North American republic are well known. The action by the colonies in definitely throwing off their allegiance to and their dependence on the mother country by the Declaration of Independence on 4 July 1776 was only the outer sign of profound historic changes of world-wide importance. With the severance of the formal links between the new 'United States' and England, there starts that strange schism in the development of American law which stands out like a main thread in the complex historical fabric of the time: on the one hand, hatred and contempt for the laws of the British Monarchy, on the other hand, a deep intuitive understanding of the time-honoured principles of English Common Law which formed the cornerstones of the colonists' own law. The American Constitution as signed on 17 September 1787 gave a final basis for all future legislation in the field of federal activities.

The small sector of American law which will be considered in this paper is founded on the concept expressed in Article I, Sec. 8, of the Constitution. The principle of patent protection is laid down there with the words that Congress shall have power 'to promote the progress of science and useful arts, by securing for limited times to authors and inventors the exclusive right to their respective writings and discoveries'.

What is the earlier history of this legal provision in one of the most important political declarations of modern times?¹ Ever since they came into existence the American colonies had been a home of forestry, hunting and agriculture. The efforts of the colonists centred upon the elementary needs of life. Food, clothing, means of transport, weapons, housing and other protection against climate and enemies formed the 'production' which was the centre of all activity. To promote useful arts and to produce necessary goods in connection therewith was an obvious step. Grants of premiums,

¹ General information about the history of American patent legislation can be found in the *Journal of the Patent Office Society*, Washington D.C., especially in the Centennial Number, July 1936, XVIII, No. 7: 'Outline of the United States Patent Office' ed. by P. J. Federico (cit. 18 JPOS), and in XXII, No. 4, April 1940/243 ff. and No. 5, May 1940/243 ff. and 352 ff.: 'Proceedings in Congress During the Years 1789 and 1790, Relating to the First Patent and Copyright Laws' (cit. 22 JPOS). Cf. also W. H. Price, *The English Patents of Monopoly* (Boston, 1913) and H. G. Fox, *Monopolies and Patents* (Ottawa, 1947).

rewards and free cultivable land had been used as encouragements for new small industries and handicrafts in the service of the colonies. By 1641 the Court of Massachusetts had already accepted a system of laws—the Body of Liberties—in which it is said ‘with us no other monopolies shall be granted or allowed as such for new inventions which are profitable for the country, and such only for short time’.² In 1672 the colony of Connecticut had formulated a very similar provision dictated by mercantilistic ideas.³ A long series of exclusive rights of a technical nature, e.g. for the production of that vital commodity, salt, was granted in Massachusetts, Connecticut, New York, Virginia and South Carolina during the second half of the 17th century. The term of validity of these monopolies varied from seven to twenty-one years. There was also the famous first patent for machinery in America which was granted in 1646 to Joseph Jenks. The Court of Boston granted this patent, though reserving to itself in perpetuity ‘the power to restrain the exportation of such manufactures and the prices of them to moderation if occasion so require’.⁴ Thereby was promoted the activity of the first iron foundry on the American continent, in Massachusetts. Small industries grew up under the protection of these oldest ‘colonial patents’. At that time village and urban craftsmen were the largest segment of the American non-agrarian population. Amongst other ‘arts’, arms manufacture was of considerable importance in Pennsylvania.⁵

After the Revolutionary War these patents of invention, which had been granted in each individual case as a result of individual decisions of the legislature of a colony, were gradually replaced by ‘state patents’ which were granted as a right to all inventors of new and useful things and no longer as specific favours. No unified legal framework or standard rules existed for the granting of these rights. Their legal application was purely local. The most important colony for machine production in the 18th century, Pennsylvania, did not grant any patent protection at all.⁶

Before going on to consider the constitutional grant to Congress to legis-

² 18 *JPOS*, p. 36.

³ *Ibid.*

⁴ *Ibid.*, pp. 38 f.

⁵ An excellent survey of artisan and craftsman life in Colonial America is given by C. Bridenbaugh, *The Colonial Craftsman* (New York, 1950). The illustrations in this book are however misleading, since they are all taken from the ‘Receuil des planches’ of the famous French *Encyclopédie* (Paris, 1762–1777).

⁶ 18 *JPOS*, p. 42.

late about technical inventions ('useful arts'), we must note another apparent source of the first federal patent legislation in America, existing side by side with the mercantilistic 'local patents' mentioned above.

The legal protection of individual literary and artistic creations is, of course, much older than the protection of technical inventions. These circumstances are similarly reflected in the development of the law in Anglo-Saxon countries. As early as 1672 a law of the colony of Massachusetts, for instance, prohibited the reprinting of printed matter without the permission of 'the owner of the copy'.⁷ Around 1780 no less than ten of the thirteen American colonies had their own copyright laws, granting to every author the exclusive right 'to the work of his intellect' as a natural right.⁸ The purpose—as the law of South Carolina (1784) expressed it, with a confusion of literary creations and technical ideas typical of this period—was 'that people should be encouraged to fulfil useful knowledges through the hope for reward, that the security of literary ownership substantially should contribute to encourage the genius, to promote useful discoveries and the general diffusion of arts and commerce'.⁹ It is, therefore, not surprising that the patent provision of the American Constitution, as it appears both in known drafts and in its final form, should regard literary authors and inventors of technical ideas as equivalent and recognizable professional groups deserving of protection by the Constitution.

In spite of all the pressing problems with which the new American republic had to contend during her early years of existence—from the cultivation of new land to the abolition of slavery—the heterogeneous convention of Philadelphia obviously considered the protection and promotion of literature and engineering as very important for the whole community. Although the final decision on the wording of the expression of principles was taken by various prominent individuals, the latter acted strongly in accordance with the conceptions and ideals of their time. Intellectual power, a great fund of knowledge and a deep conviction of the value of man distinguished those who created these laws. But there was no uniformity of approach amongst them. Their opinions about the power which the legislature of the country should have varied according to where in the colonies they had been born or brought up. In 1783 the Continental

⁷ *Ibid.*, p. 36.

⁸ *Ibid.*

⁹ *Ibid.*, pp. 43 f.

Congress had already accepted a resolution to recommend all states to secure copyright for authors.¹⁰ Of the debates of the Constitutional Convention in Philadelphia in the spring of 1787 no minutes were kept and in many respects its work must therefore be reconstructed from the diaries of participants. Such a diary was kept by one of the most important 'Founding Fathers', James Madison.¹¹ Madison has been called 'the master builder of the Constitution'¹² and it has been said that he had his hand in nearly every legislative project.¹³ This was certainly true of the preparation of the Patent Clause of the Constitution.

As is well known, the main purpose of the Convention was to create a new federal state system with a balance of powers between central and local authorities and a tripartite government. These principles had been described theoretically in Europe by Locke and by Montesquieu. The protection of authors and inventors was clearly considered to be purely a matter for federal legislation. In August 1787 both James Madison, thirty-six year old representative of Virginia, and Charles Pinckney, thirty year old representative of South Carolina¹⁴, each submitted a proposal for a constitutional provision to protect literary works and inventions. Both these men were typical aristocrats of the American South—conservative and friends of a strong unified government. Madison suggested the following wording in the Constitution:

To secure to literary authors their copy rights for a limited time. To encourage by premiums and provisions, the advance of useful knowledge and discoveries.¹⁵

Pinckney's proposal was: 'To grant patents for useful inventions. To secure to authors exclusive rights for a certain time.'¹⁶ Pinckney's own state, only three years earlier, had issued its first, carefully prepared, common copyright and patent law, according to which inventors of useful machines could receive the same fourteen-years' exclusive right for their inventions as

¹⁰ *Ibid.*, p. 56.

¹¹ Notes of Debates in the Federal Convention of 1787, published in M. Farrand, *The Records of the Federal Convention of 1787* and in *Writings of James Madison*, ed. by G. Hunt (1900–1910), III and IV. *Documents on the Formation of the Union* (Washington, 1927) should be studied, but were not available in Sweden.

¹² M. Farrand, *The Framing of the Constitution of the U.S.* (New Haven, 1913), p. 196.

¹³ *Dictionary of American Biography*, ed. by D. Malone (New York, 1933), XII, 184 ff.

¹⁴ 18 *JPOS*, p. 56.

¹⁵ *Ibid.*

¹⁶ *Ibid.*

authors for their books. In order to get such an exclusive right an administrative decision of the local parliament was still apparently necessary in each individual case. In an earlier report Madison had already asked the Continental Congress for unified legal regulation of literary property in all states.¹⁷

These were primarily questions of a purely legal nature, especially of constitutional law. But precisely at this time, the granting of patents for inventions was also proving to be a very real problem. Two American inventors, James Rumsey and John Fitch, had attracted the attention of responsible leaders of the republic. Both men believed that they had invented steamships of practical use and they tried in numerous ways to secure recommendations and funds for their respective projects.¹⁸ In the economic circumstances of the new nation, the development of navigable waterways, for instance along the Hudson, the Potomac, the Ohio and the Mississippi rivers, was a key to the further development of the country. It is easy therefore to understand that the application of new ideas from England concerning the use of engines to drive boats was of outstanding interest in contemporary America. Rumsey and Fitch wrote to George Washington, to the various state legislatures as well as to the American Philosophical Society of Philadelphia, a centre of progress for all the 'useful arts'. Rumsay filed patents, in several states including Virginia, Maryland, Pennsylvania, New York, New Jersey, North Carolina and Georgia. In the first three of these Rumsay was granted exclusive rights, in the remaining four he was unsuccessful. Fitch, however, in 1785 submitted a memorandum to the American Philosophical Society in Philadelphia for publication in its famous *Transactions*.¹⁹ In the same year he presented to Congress in New York drawings and models for a steam-powered ship and asked for a congressional guarantee of the proceeds of the sale of certain maps which he had drawn and printed in order to cover his costs for acquiring patent rights. Meanwhile Rumsay had already succeeded in May 1785 in securing a resolution of Congress making a free grant to him of 30,000 acres of land, west of the Ohio river, on condition that he proved before May 1786

¹⁷ *Ibid.* or J. Brant, *James Madison, Father of the Constitution 1787-1800* (Indianapolis, 1950), p. 132.

¹⁸ Details about the history of steamboats in the United States, cf. F. D. Prager, 'The Steamboat Pioneers before the Founding Fathers', 37 *Journal of the Patent Office Society*, pp. 486 ff. (*cit.* 37 *JPOS*); J. T. Flexner, *Steamboats Come True* (1944).

¹⁹ 37 *JPOS*, p. 504.

that a ship of his construction could travel against the stream of that river for six days at the rate of 50 miles per day.²⁰ In the event, Rumsay did not become an owner of part of the territory of the United States of America since he failed to fulfil the given conditions. Legally this kind of reward was a bounty, being a premium paid by the government to encourage some branch of production. The rivalry between the two inventors continued with a demonstration of Fitch's steamboat. This was given to the members of the Constitutional Convention and took place on the Delaware river on 20 August 1787.²¹ This demonstration took place at the precise time when Madison and Pinckney were working on the patent clause of the federal constitution. A petition from Fitch on this matter had also been sent to Madison on 11 August. On 5 September a special committee presented the draft of a patent clause to the Convention and this assembly unanimously accepted it the same day *nem. con.*²² The delegates to the Convention thus seem to have felt the need for a unified and common federal law relating to inventions.

The Constitution, however, needed to be ratified by nine states in order to become law and not all states were equally enthusiastic about the different provisions and clauses. Here, once more, Madison's activities were outstanding and decisive, albeit well-nigh anonymous. Between October 1787 and the beginning of 1788 there had appeared, under the name of 'Publius', the eighty-five articles, which came to be known as the *Federalist* and which systematically explained and defended the new Constitution.²³ The forty-third in this series, probably written by Madison in January 1788, represents the only existing contemporary comment on the patent clause in the federal Constitution. The problem was described in the following words:

The utility of this power will scarcely be questioned. The copyright of authors has been solemnly adjudged, in Great Britain, to be a right of common law. The right to useful inventions seems with equal reason to belong to the inventors. The public good fully coincides in both cases with the claims of individuals. The States cannot separately make effectual provision for either of the cases, and most of them have anticipated the decision of this point, by laws passed at the instance of Congress.

²⁰ *Ibid.*, p. 499, or *Journals of the Continental Congress*, L.C. ed. (Washington, 1904–1937), XXVIII, 349 f.

²¹ 37 *JPOS*, pp. 517 and 518, note 144.

²² 18 *JPOS*, p. 57.

²³ *The Federalist, A Commentary on the Constitution of the United States*, reprint (New York, 1937), No. 43, 278 f.; M. Farrand, *op. cit.*, p. 179.

The declaration illustrates the ambivalent attitude towards English laws, mentioned above, which the creators of the American Constitution must have felt. The same men who in 1776 turned their back in disgust on the mother country and her institutions admitted shortly afterwards, and rightly, the unsurpassed wisdom of the English Common Law in most fields.

When North Carolina had ratified the Constitution at the end of 1789, the work on the Constitution was completed. On 8 January 1790 the first President of the United States made a speech in Congress in which he said:

The advancement of agriculture, commerce, and manufactures, by all proper means, will not, I trust, need recommendation, but I cannot forbear intimating to you the expediency of giving effectual encouragement as well to the introduction of new and useful inventions from abroad, as to the exertions of skill and genius in producing them at home. . . .²⁴

The encouragement of inventive work had obviously found the sympathy of the president. Washington justified his attitude partly with the old mercantilistic concept that the importation of new methods from abroad should be encouraged (by exclusive rights), but in view of the backwardness of economic and technological development in the United States at that time, he was more or less compelled to recommend any means by which the great engineering innovations, especially steampower and the textile machinery newly developed in England, could be introduced into the new republic. Congress reacted speedily to the appeal of the President. A particular reason for this was that petitions for the grant of copyrights and patents, having still to be determined by individual decisions, constituted a heavy burden on the work of Congress. A series of such petitions were just awaiting action, among them some concerning the new steamboat inventions of John Fitch. Two committees were set up. Appreciating correctly that literary property and technical inventions can only with difficulty be treated in the same legal enactment, two bills were prepared which gradually led to two different laws.²⁵

The Patent Committee consisted of Edmund Burke from South Carolina, Benjamin Huntington from Connecticut and Lambert Cadwallader from New Jersey.²⁶

²⁴ Cit. 22 *JPOS*, pp. 253 f.

²⁵ *Ibid.*, pp. 259 ff.

²⁶ Cf. *Biographical Directory of the American Congress, 1774–1927*, Washington, quoted in 22 *JPOS*, pp. 246, 259.

The patent law bill was debated early in 1790 in both Houses of Congress. The bill was accepted by the House of Representatives and by the Senate, and the law was signed by the President on 10 April 1790. With this signature the first federal patent law of the United States had been enacted.²⁷

It is not known what role Alexander Hamilton played in preparing the ground for federal patent legislation. Hamilton wrote about fifty of the eighty-five numbers of the *Federalist*²⁸ as well as the very remarkable *Report on the Subject of Manufactures*.²⁹ On 15 January 1790 Hamilton had been instructed by the House of Representatives to study and report on means of promoting manufactures in the United States; he submitted his report on 5 December 1791. Hamilton had therefore received his instructions during the course of the patent debate in Congress, but he probably did not have even a draft of his report ready when the law went into force in April 1790. As a member of the government he did not take part personally in congressional debates.

Little has been said as yet of another very important figure in this field: Thomas Jefferson. Jefferson, the author of the Declaration of Independence, was from 1784 to 1789 American Minister to France and could not therefore personally influence the convention of the Constitution. He had however pursued a lively exchange of ideas in correspondence with the foremost architects of the Constitution, Hamilton, Madison, Washington and Franklin. When he returned to America in 1789, he was immediately called on to carry out essential government work, amongst which was to transform the new federal patent law into a practical working system. A knowledge of Jefferson's opinions on the patent system and inventions is therefore essential for an understanding of American legal history. A study of his views on these questions as expressed in letters and articles or as made manifest in law bills shows that he experienced some radical changes of opinion. Consequently it is not difficult to find quotations from his writings to support the most varying interpretations of patents and inventions. In defence of his changing attitudes during the years it must be remembered that in the course of his life—as private landowner in Virginia, as a diplomat

²⁷ On 31 May 1790, the first American Copyright Act was enacted.

²⁸ See above, p. 132, note 23.

²⁹ Alexander Hamilton, *Papers on Public Credit, Commerce and Finance*, ed. by S. McKee Jr. (New York, 1934), pp. 175 ff., concerning inventions, especially pp. 240, 243, 275.

in France, as a member of the Government and as President of the United States—Jefferson saw the problem from different angles and in a different light. His letters are a mine of information on many matters.³⁰ His correspondence from 1784 to 1789—when he was American ambassador in France—illustrates the way in which he reported to America about various new inventions in Europe, especially in England.³¹ In connection with a visit to England Jefferson wrote admiringly about the application of the steam engine to mills. He also had personal talks in Paris with Matthew Boulton, the partner of the British inventor, James Watt.

One of Jefferson's earliest remarks about the legal protection of inventions is probably that in a letter dated 9 August 1787 to the French inventor, L'Hommande, in which he politely refused to accept a French invention on behalf of the United States. Jefferson stated sceptically that 'though the interposition of government in matters of invention has its use, yet it is in practice so inseparable from abuse, that they think it better not meddle with it'.³² However, in this particular context, the most interesting of the many letters which Jefferson wrote to his countrymen at home, are those which he addressed to James Madison who, as we have seen, was jointly responsible for the patent clause of the Constitution.³³ In a comprehensive letter of 20 December 1787 Jefferson sent to Madison from Paris a kind of review of the new constitutional draft, the shaping of which he could only observe from the distant French scene. Among the shortcomings he found in the document was 'the omission of a bill of rights providing clearly and without the aid of sophisms for freedom of religion, freedom of press, protection against standing armies, restriction against monopolies . . .'.³⁴ Jefferson considered that the American people were entitled to some statement on these matters notwithstanding the discord among the states. In a letter to Madison dated 31 July 1788 Jefferson returned to the question of mono-

³⁰ The letters of Thomas Jefferson are collected in different editions, the two most important ones being by H. A. Washington, *The Writings of Thomas Jefferson* (New York, 1861), in 9 volumes (*cit.* Wash. ed.), and by P. L. Ford, *The Works of Thomas Jefferson* (New York, 1904–1905), in 12 volumes (*cit.* Ford ed.).

³¹ Cf. Jefferson, *Report of a Conference with the Count de Vergennes* (on January 2, 1786), pp. 180 ff., and autobiography in *The Complete Jefferson*, arranged and assembled by S. K. Padover (New York, 1943), chapter XXIV, 1161.

³² Wash. ed. II, 236 f.

³³ For the relations between Jefferson and Madison, see, e.g., I. Brant, *James Madison* (Indianapolis, 1950), chapter XXVII.

³⁴ Ford ed. V, 371.

poly and patents. He expressed his satisfaction that nine states had already accepted the federal constitution which he described as 'a good canvas on which some strokes only want retouching'; he noted, for instance, the lack of a general declaration which *inter alia* would do away with monopolies. He went on:

The saying that there shall be no monopolies lessens the incitements to ingenuity, which spurred on by the hope of a monopoly for a limited time, as of 14 years, but the benefit even of limited monopolies, is too doubtful to be opposed to that of their general suppression.³⁵

A year later, on 28 August 1789, Jefferson, still in Paris, wrote once more to Madison about the new federal constitution. He wrote that he liked the declaration of rights as far as it went, but 'I should have been for going further'. He suggested some changes and additions about freedom of speech, the jury system, loss of liberty, and similar matters. As a new section 9 in the constitution he proposed: 'Monopolies may be allowed to persons for their own productions in literature and their own inventions in the arts, for a term not exceeding - - - years but for no longer term and no other purpose.'³⁶ When this letter was written many of the fateful events of the French Revolution had just happened, in some of which Jefferson's personal friend Marquis Motier de La Fayette had been involved. On 26 September 1789 Jefferson bade farewell to France by writing to the Secretary of Finance, Necker, with the conciliatory gesture that he intended to ask the American government to send 'salted provisions', at half the price of fresh meat, to the starving masses of Paris.³⁷

In October 1789 he returned to America and in December was appointed Secretary of State in the new government. The five years he spent in France, where he had witnessed the gradual growth of tension between the *ancien régime* and its enemies, finally erupting into revolution, did not pass without leaving a deep impression upon him.

In April 1790, as already mentioned, the first patent law bill was accepted. Its title was the 'Law to promote the progress of the useful arts'.³⁸ The importance attached to the law by Congress is evident from the fact that

³⁵ Ford ed. V. 428.

³⁶ Ford. ed. V, 493.

³⁷ Wash. ed. III, 119 f.

³⁸ The full text of the Patent Act of 1790 is found in *The Public Statutes at Large of the United States of America*, I (Boston, 1845), 109 ff. and reprinted in 14 JPOS, pp. 237 ff. as well as in 22 JPOS, pp. 283 ff.

patent applications had to be filed with at least one of the following: the Secretary of State (then the newly appointed Thomas Jefferson), the Secretary of War (Henry Knox) or the Attorney General (Edmund Randolph). According to the law 'any useful art, manufacture, engine, machine or device or improvement therein' was patentable if it had not been known or used before. The appraisal of 'novelty' was left entirely in the hands of the three ministers as was also the decision on a further condition: that the invention according to their judgment was 'sufficiently useful and important'. Consequently usefulness and importance were judged wholly from the point of the momentary needs of the administration. The exclusive patent right for inventions covered permission to make, build, use and offer them for sale. The monopoly right was granted for 14 years and was valid in the entire territory of the republic, though apparently the state patents which had previously caused so much trouble were not abolished at the same time. The requirements for the specification of patents and for drawings and models were severe. They should be exact enough to make it 'possible for a workman or other person who was expert in the field, to make, build or use the same'. The purpose of such an application was that after the end of the life of a patent, the public should have the entire and free benefit of the invention described. It was typical of the evolution of American patent law that in this first federal law of 1790, a patent was only to be granted to 'the original and first inventor', verified by oath. This was in exact agreement with section 6 of the English Statute of Monopolies of 166 years earlier.

American patents were fairly cheap to acquire at that time: filing fee 50 cents, sealing fee \$ 2, big seal \$ 1, description 10 cents per page (100 words). Patents were provided with the seal of the President and registered in a ledger in the Department of State.

The first patent commission to operate was dominated by Thomas Jefferson by dint of his great personal interest in science and engineering. Jefferson had some special knowledge of physics, mechanics, astronomy, botany, anatomy and architecture. He had also reading knowledge of at least five languages.³⁹ Jefferson has often been described as a great inventor himself. In a wider scope, however, this view can hardly be sustained. Jef-

³⁹ E. T. Martin, *Thomas Jefferson: Scientist* (New York, 1952); Holland Thompson, *The Age of Invention: A Chronicle of Mechanical Conquest* (New Haven, 1921); Browne, *Thomas Jefferson and the Scientific Trends of His Time*, pp. 399 ff.

person was an 'inventor' in a sense similar to that in which Goethe or Linné were 'inventors', i.e. he had the conviction of the 18th century that science should be applied for the good of all people, and the interest of the alert and curious man in all the basic discoveries which were made during his life time, especially in France and England. The most advanced aspect of Jefferson's engineering knowledge was probably his acquaintance with the structure of buildings. He himself partly designed his elegant home in Monticello, buildings belonging to the University of Virginia in Charlottesville, and the Capitol in Richmond, Va. For his well-known mechanical gadgets, however, he never applied for patent protection. A short time after the first patent law came into force, he wrote in a letter from New York to Benjamin Vaughan on 27 June 1790—apparently with some surprise—that the new law 'has given a spring to invention beyond my conception' and that he now felt familiar with inventions, 'being an instrument in granting the patents'. He continues in the same letter that 'many of them indeed are trifling, but there are some of great consequence, which have been proved by practice, and others which, if they stand the same proof, will produce great effect'.⁴⁰ It would be interesting to trace further the attitude of Jefferson towards the patent system, as revealed in some comprehensive letters from him to a number of American inventors during the years 1803, 1807, 1813⁴¹ and especially 1814.⁴² Quotation from them, however, would unduly swell the length of this paper. It will suffice to mention that Jefferson, on the basis of his practical experiences with the first patent law, had already in February 1791 finished a draft for a law bill which led two years later to the second patent law of the United States.⁴³ This law changed the requirements concerning proof of novelty; instead of leaving this to the members of the official patent commission, it was required that printed material should be cited by those opposing the grant. This system did not work, however, and a law of a more modern type was enacted in 1836.

⁴⁰ Wash. ed. III, 158.

⁴¹ In a remarkable letter of 13 August, 1813, from Monticello, addressed to the inventor Isaac M'Pherson, Jefferson explains in nine pages the question of what we today call patent infringement; he then develops an extensive enquiry into the novelty of the inventions of Oliver Evans of elevators, conveyors and 'hopper-boys' by citing relevant engineering literature, especially from France, in a manner similar to that in which Official Actions are now written by the United States Patent Office (Wash. ed. VI, 175 ff.).

⁴² Wash. ed. VI, 175 ff., 295, 297 ff., 372.

⁴³ The bill 'To promote the Progress of the useful Arts' of 7 February 1791 is reproduced in *The Complete Jefferson*, assembled by S. K. Padover (New York, 1943), pp. 995 ff.

III

France

In France as in America the historical precedents for the first modern patent legislation, a law passed in the same year as in the United States, may be traced back to the mid-17th century.⁴⁴ The French monarchy of the time was closely concerned with the effective maintenance of corporate and guild organisation. The famous products and artistic work of contemporary France were ordered and paid for by the royal family and their immediate associates. These products, be they silk fabrics or tapestries, glassware or porcelain, leather goods or hand weapons, reflected the luxury and splendour of the upper classes. The French guilds had important centres for production and sale; the silk weaving mills in Lyon and Tours, for instance, were large even by modern standards, and at such places engineering and 'useful arts' were practised. At the same time basic sciences were developed under the supervision of the French Academy of Sciences.⁴⁵ In both institutions exclusive privileges for technical methods and products were known. The character of these privileges, however, was very different from the patent monopoly according to modern definitions. In the first place, the authority to grant such exclusive rights belonged to the French king personally or to his delegate. It depended on his judgment or whim, whether and by what means a new invention should be promoted. Secondly, the monopoly which was granted to a guild was, from the beginning, a collective monopoly which could not be enjoyed by an individual inventor but only by a corporation as an economic unit. At the same time, however, some prudent persons succeeded in securing individual patents of inventions for a number of years by petitioning the Academy of Sciences, some prominent minister or the king personally; they were granted in return for gratuities for a varying period of time. Some famous examples include the patent of 1649, given to the philosopher Blaise Pascal for a calculating

⁴⁴ Valuable details about the history of industrial property rights in France can be found in E. Levasseur, *Histoire des Classes Ouvrières et de l'Industrie en France avant 1789*, I, II (Paris, 1900–1901), cit. Levasseur (A); E. Levasseur, *Histoire des Classes Ouvrières en France depuis 1789 jusqu'à nos jours*, I, II (Paris, 1867), cit. Levasseur (B); E. Levasseur, *Histoire du Commerce de la France* (Paris, 1911); E. Martin-Saint-Léon, *Histoire des Corporations de Métiers* (Paris, 1941); Alexis de Tocqueville, *The Old Regime and the French Revolution*, translated by St. Gilbert (New York, 1955).

⁴⁵ Levasseur (A), II.

machine; and those of 1657 and 1675 to the Dutch physicist Christian Huygens for a pendulum clock and for a miniature spiral spring pendulum respectively, which made it possible to manufacture watches.⁴⁶

At about this time Louis XIV's celebrated minister, Jean Baptiste Colbert, turned his mind to the advantages and disadvantages of the existing method of granting exclusive privileges. In about 1670 he made some very critical remarks about applications for certain privileges such as that for the manufacture of crêpe fabrics, normally imported from Italy, and for the manufacture of porcelain.⁴⁷ The grant of a monopoly, by royal grace, to an inventor or importer, was, in many instances, based not on technical considerations but would often comprise both artistic products and purely commercial activities. And the borderline was far from clear.

The emigration of many skilled craftsmen to England, Holland and elsewhere, following upon the revocation of the Edict of Nantes, had, as is well-known, a most harmful effect on the development of French industry. In 1699 the Academy of Sciences in Paris, for which had been chosen the motto *Invenit et perficit*, introduced a provision in its rules prescribing that, on the order of the king, it should issue certificates stating whether certain machines were new and useful; it was also to invite 'recognised inventors' to deposit models with the Academy. These were obviously functions of the type normally associated with a Patent Office. Even 'patent lawyers', to assist inventors in exploiting inventions, seem to have existed in Paris around 1690.

The first legislation in the field of industrial property rights in France refers to the question of particular designs. Statutes of 1712 and 1744 relate to *les marques* and *les plombs* which became compulsory for all goods produced by craftsmen employed in factories licensed by royal privilege.⁴⁸ The idea that a specific way of working textiles, e.g. spinning, weaving and embroidering silk, could be a 'property' worth protecting by law was an entirely new concept in France. Regulations, aimed primarily at identifying origin, indicating the use of certain designs and techniques, and controlling both quality and price were administered first by special local officials, and

⁴⁶ *Lettres, Instructions et Mémoires de Colbert*, II: 2 (Paris 1863), 694 f., and Appendix, pp. 751 ff., 'Edit du Roy'.

⁴⁷ *Ibid.*, p. 290.

⁴⁸ F. D. Prager, 'A History of Intellectual Property from 1545 to 1787', 26 *JPOS* (1944), pp. 728 ff. (*cit.* Prager).

later by state officials. The marking of goods, however much of a nuisance in practice, was held to be necessary not only in the various branches of the French textile industry, in the weaving of linen, silk and woollen cloth, as well as in the cutting, sewing and dyeing of fabrics, but also in the manufacture of paper, glass, porcelain and furniture.

A further circumstance which affected the development of legislation for industrial property was the growth of an extensive literature on arts, crafts and natural sciences which began during the French Enlightenment. It inspired inventors and industrial experts by thorough and detailed descriptions of the arts of all Europe. Denis Diderot and Jean Le Rond D'Alembert demonstrated the progress of industries and applied sciences by editing the famous twenty-eight folio volumes of the great Encyclopaedia between 1751 and 1772, a publication which was not at all to the liking of the guilds anxious to conserve old and approved methods of work.⁴⁹ Diderot, previously imprisoned for his literary attacks on the monarchy and the church, contributed further to the new direction of industrial property law in France by a remarkable paper entitled *Lettre sur le commerce de la librairie* of 1767.⁵⁰ Its publication was caused by a dispute over the right to publish the works of La Fontaine. Both the author's grandchildren and his publisher had apparently received exclusive rights, the publishing firm by direct purchase from the author, the grandchildren by right of a royal favour. The guild of publishers engaged Diderot, the enemy of monopolies, to defend their case; they hoped thereby to ensure a general attack on the arbitrary actions of the Crown as well as on all forms of privilege. Diderot, however, wrote a general survey of the privilege system as practised by the French publishers through the years, and fashioned his complaint into a constructive and positive approach which, according to Prager, led eventually to the concept of intellectual property as a part of a free society.⁵¹

At the same time, the Physiocrats under the leadership of Quesnay attacked the governing regime. Part of their programme was to ask for a general legal protection of property. By appointing a leading physiocrat and encyclopaedist, Jacques Turgot, to be Comptroller General in 1774, the new king Louis XVI pushed the development ahead. Two years later Turgot

⁴⁹ F. Neumeyer, 'Amiral F. H. af Chapmans Architectura Navalis Mercatoria', *Tidskrift i Sjöväsendet*, Nr. 2 (Karlskrona, 1943), 25.

⁵⁰ The paper was rediscovered in 1861. The authorship of Diderot is not certain.

⁵¹ Prager, p. 733.

issued his famous six Edicts, of which one abolished all guilds, as a matter of principle. At the same time he said:

We consequently wish to abrogate these arbitrary institutions, which . . . retard the progress of the arts by reason of the manifold difficulties which inventors meet, different communities disputing their right to put into practice discoveries to which the communities have in no way contributed.⁵²

In Turgot France had a political leader of visionary power who sought to liberate the economy and the country's productive resources from its medieval fetters by force, but—'the measure was revolutionary in a non-revolutionary state'.⁵³ Turgot's friends, for instance, Condorcet and Voltaire, greeted his actions with enthusiasm (Voltaire wrote to Turgot in January 1776: 'vous faites naître un beau siècle dont je ne verrai que la première aurore . . .'),⁵⁴ but in the Parliaments, their reception was very different. The guild associations were declared to be links in a chain which held together the French monarchy. In August of the same year Turgot was forced to retire and the craft guilds were re-established, largely by Turgot's successor, Necker. Despite this seeming total failure to advance the new ideas, the old corporate class structure and its 'links' were breaking up. Two years later, it was noted in a memorandum edited by the powerful *Bureau de Commerce*, the central organization of the French government for the control of industrial regulation, that:

The grant of the title of royal factory is an honour and a certificate of excellence in manufacture which works to the detriment of other manufacturers. It is a judgment of the administration which is not authorized to judge in this way. It is necessarily a judgment which is unjust, because it is a permanent judgment and the state of things is changing. . . .⁵⁵

However, in spite of this criticism, the memorandum came to the conclusion, that it would be better, for reasons of economy, that industry should be encouraged by royal privileges rather than by monetary grants or monopolies accompanied by gratuities in cash, to individual inventors.

During the preceding year, 1777, two statutes had been passed which were relevant to these questions. One regulated the copyright laws, the other the

⁵² For the wording of the edict in question (*Edit du Roi portant Suppression des Jurandes*), see Levasseur (A), II, Appendix, 'Piece C', 867 ff.

⁵³ Eli Heckscher, *Merkantilismen* (Stockholm, 1931), I, 188.

⁵⁴ Quoted Levasseur (A), II, 629, note 1.

⁵⁵ *Ibid.*, p. 665.

right of painters and sculptors to their creations. In view of the importance of the creative arts in France at that time, and of the employment of domestic and foreign craftsmen, it is readily understandable that the regulation of artistic property rights should have been regarded as necessary for the whole country, even though it was often merely a question of luxuries for the Royal family and the upper classes.

Certain other relevant developments in the French economy must now be briefly considered. It is well known that, about 1770, the new technique of making cotton fabrics was introduced into France from England.⁵⁶ Manufactured goods as well as various sorts of inventions for weaving, finishing and dyeing cotton fabrics arrived from England; technicians and other experts also appeared, sometimes as entrepreneurs in the French centres of industry and commerce. The French manufacturers tried to resist this movement, but the government and the king encouraged the new English methods in spite of the obstacles presented by French guild regulations. The new English spinning machinery gradually came into use, and drawings or models of inventions, generally protected by patents in England, were offered to the French. The king rewarded a Mr. Milne with 60,000 livres gratuity and an annual pension of 6,000 livres for the importation of a new English cotton spinning machine.⁵⁷ The Irishman Holker was permitted to start a big French factory for the manufacture of woollen cloth, and was granted by Louis XVI 90,000 livres and the title of inspector general of foreign factories. Holker published instructions about the right way of manufacturing English fabrics, founded a weaving school which was run by an Englishman and introduced English workers into the French factories.⁵⁸ Other new technical methods, for instance the manufacture of English

⁵⁶ The history of the different stages in the machine manufacture of fabrics in England, and the way these basic inventions penetrated other countries is most important for the history of patent legislation and engineering in Europe and in the United States. Since machinery for cotton spinning, weaving and carding was patented in England but not in other countries, interesting problems arose when the British inventors and their manufacturers exported such machinery and the relevant 'know-how'. In France the British importers met an old-fashioned and arbitrary system of patent privileges; the king determined both the extent and the duration of the privilege as well as the compensation paid for it. Essential facts concerning the impact upon France of advanced British technology in cotton manufacture in the 18th century can be found in A. P. Wadsworth & J. de Lacy Mann, *The Cotton Trade and Industrial Lancashire, 1600-1780* (Manchester, 1931), especially books II and V, and in E. Levasseur (B), I, 70 ff.; cf. also W. O. Henderson, *Britain and Industrial Europe 1750-1850* (Liverpool, 1954).

⁵⁷ Levasseur (A), II, 540.

⁵⁸ *Ibid.* and Wadsworth & Mann, *op. cit.*, pp. 195 ff. and 503 ff.

pottery, were developed in France. The French government paid rewards, either as lump sums or extending over several years, to the foreign innovators as well as granting them exclusive privileges.

Patents of limited duration are today granted in France and certain other countries, under the name of 'patents of importation', for inventions of which it is demanded that they should be new in the country into which they are imported. The giving of special weight to such importation of new technical devices derived from the well-known invention of the spinning machine as well as from that of the power loom (patented in England respectively by Richard Arkwright in 1769 and Edmund Cartwright in 1785). Arkwright's machines were designed to be driven by horse- or water-power. English inventors had already had access for over 150 years, thanks to the Statute of Monopolies of 1624, to a clear, unified system of patent protection, and they made use of this, with varying success, in many important matters.

This English 'invasion' of France during the 1770's and 1780's touched a very sensitive point in the national self-confidence of the French and helped to stimulate the eventual appearance of French legislation for patents of invention.

As late as 1791 numerous complaints about foreign competition were heard from various centres of industry, e.g. Rouen, Reims, Rennes and Lyon; the French government were forced to introduce strong protective tariffs.⁵⁹ In the 1780's there were disputes over different patent rights. Three Frenchmen had received, in 1784, an exclusive patent for 12 years enabling them to erect factories for the continuous production of yarn and of muslin and other cotton fabrics. In the same year another Frenchman protested against this patent, which covered the mechanical manufacture of cotton by water power, having himself been granted a 15 years' patent for a machine to spin wool and cotton. In 1787 another person wanted to build a factory equipped with spinning machines of the same type in Lyon and demanded a privilege in accordance with Necker's edict of 1776, and this right was granted to him.⁶⁰

The lack of a national unified patent law became more and more evident as the pace of industrialization quickened. The willingness of French authorities and of politicians gradually to bring to an end the old economic

⁵⁹ Levasseur (B), I, 14 f.

⁶⁰ Levasseur (A), II, 666.

organization based on the guilds and to reform obvious anomalies was in the long run inadequate. Moreover, some of the royal edicts and regulations were disregarded in the country districts and deliberate opposition to government plans was apparent.

With the revolution of 1789 there was created in France an entirely new basis, both ideological and practical, for the system of patent protection of inventive work. It will be sufficient to recall the wording of the 'Declaration of Human Rights', where it could be read: 'laws should be clear, exact and alike for all citizens'. Although, as we have seen, attempts had been made since 1712 to regulate artistic and industrial property rights by statute, it was through the legislation of the Revolution that for the first time the unlimited personal power of the absolute monarchy, in this as in other fields, was removed and the right of the individual to his own intellectual creations received its recognized place in society.

The Constituent Assembly of 1789 had a great amount of legislative work to do in order to remove the worst anomalies of the backward *ancien régime*. Among other matters, the organization of the law courts was entirely changed by introducing, for instance, the jury system in criminal cases, public court sessions and the right of a defendant to have recourse to a lawyer.⁶¹ It was in the course of this great work of legal reform that there came to be considered the problems of the royal system of privileges and the patents of inventions.

The events which ostensibly resulted in the first national patent law, seem to have been a proposal by the chamber of commerce of the province of Normandy in the summer of 1790 and a petition by inventors sent to the parliament in Paris. When these documents reached the Assembly they were referred to the Committee of Agriculture and Commerce for the drafting of a law bill. The Committee prepared a report which was submitted to the Assembly on 30 December 1790.⁶² The contents of this report and its author are important both in the history of ideas and in the history of law. The title of the report was *Sur la propriété des auteurs de nouvelles découvertes & inventions en tout genre d'industrie*, and it was written by Marquis Stanislas de Boufflers.⁶³ Boufflers was a romantic personality. He

⁶¹ Cf. A. Söderhjem, *Den stora Revolutionen* (Stockholm, 1927), I, 116 ff., and *Kulturförhållandena under Franska Revolutionen* (Helsingfors, 1903), p. 123.

⁶² The report is an appendix to *Procès-Verbal de l'Assemblée Nationale*, imprimé par son Ordre, Onzième Livraison, Tome XLI (Paris, 1791).

⁶³ b. 1738, d. 1815.

was the son of a lady of the court of the Polish emigrant king Stanislas Leczinski, resident at Lunéville in France. As a young man Boufflers had been an officer and had gone to Poland in 1771. In 1785 he became governor of the French West African colony of Senegal. He was also a poet, and in 1788 he was appointed, on the merits of his poetry, to membership of the French Academy. The following year he was elected as deputy to the National Assembly, for the nobility of the district of Nancy, and was entrusted with enquiring into various legal projects, in which context he had to consider the law of inventions.⁶⁴ Obviously Boufflers had no special qualifications for the task of preparing the draft of a patent law. Nevertheless his report is a most interesting document.

If the report is stripped of the rhetorical phraseology of the times, it can be seen to deal mainly with three matters: the backwardness of French industry, the English penetration of the French economy, and the wish to improve the standard of living of French industrial workers. The two first were obviously of importance to the national economy; the third represented the enlightened social viewpoint which Boufflers and other representatives of the nobility were adopting at this time, the autumn of 1790.

Boufflers based his report on the ~~concept~~^{idea} that the real property of man is his ideas. The invention is the source of handicrafts and of property. There is a natural property (*concession de la nature*) and a property through the community (*concession de la société*). The inventor asks according to this conception only that the community (*le corps social*) shall guarantee this property. The community can hereby profit, but cannot lose anything. Since the future advantages of an invention are distant as well as uncertain, the inventor must provide convincing evidence as a pledge that he is willing to keep his 'contract', that he will indemnify the community for services received and that he will offer his country the first products (*les prémices*) of his invention.

Boufflers considered this settlement to be a *pacte raisonnable*. He then severely criticised the prevailing conditions in France, attacking the opposition to progress represented by guild regulations. He attacked the excessive regard for routine, as well as the demands of the French treasury,

⁶⁴ Before the terror Boufflers emigrated to Prussia, where Prince Henry made him a member of the Prussian Academy of Berlin and gave him a large area of uncultivated land in Poland. In 1800 Boufflers returned to France and in 1803 was restored as a member of the French Academy. During the restoration he became librarian of the *Librairie Mazarin* in Paris, where he died in 1815.

le fisc, which, he said, armed with its great powers, destroys everything, degrades morals, laws and genius alike; isolates, oppresses and perverts all useful professions. He described how the inventor had humbly to submit his discoveries to the 'arrogant Aristarchos', meaning presumably the examiners of the French Academy of Sciences. These men, though they might be wise and good, were too often, he thought, biased, being sometimes agents of the treasury or members of a guild which was afraid of direct competition in its work, as a result of new inventions. 'C'est au propriétaire de cette idée à la faire valoir; il en a le droit, il ne lui faut plus que la sécurité. Protégez-le donc, et ne le payez point.'⁶⁵ What Boufflers demanded was a general legal protection of the ideas of the inventor and not arbitrary alms. At this point he introduced two new principles of a revolutionary character for France: (1) equality of all citizens before a unified national law of inventions, and (2) the granting to the inventor of the right to secure the rewards which flow from a successful invention, instead of the state giving a reward to the inventor.

Here Boufflers had much to say about English arrangements: the existence for the last 150 years of wise legislation concerning inventions, the enlightened practices represented by the right to go to law, freedom of the press and the person, the sympathetic attitude of the government to new commercial and industrial activities. On the other hand, as he saw it, there was the dominating position of the English in French markets, where French managers of royal factories were said to be agents for English goods. Boufflers claimed that of every thousand French privileges for inventions nine-tenths were granted to foreigners. He likened these activities of the English to those of a king who attacks his enemies with their own soldiers. Behind this rhetoric lay the urgent and vigorous desire to persuade the Assembly to follow a path entirely foreign to French tradition.

At the end of his long report—after having decided that the English should be imitated in order to be surpassed—Boufflers wrote:

It was not out of an empty and exaggerated conscientiousness that the proud and wise Americans, those dignified friends of all liberty, accepted in their federal Constitution the principles of English industrial legislation as the safest remedy to secure the freedom and prosperity of their own industry. Cannot this mania to imitate, of which we have so often and so justly been accused, be for once applied to something other than simple and superficial things? Must it be ended just at the moment when imitation

⁶⁵ P. 14 of the original French report (see above, note 62).

is becoming sensible? To imitate everything is childish, to refuse everything is foolish. In short, as the Americans, the most recent allies of liberty, have accepted English patent law, then legislation in the same spirit cannot be unsuitable in contemporary France.⁶⁸

Behind this reference to America lay various sources of influence, direct and indirect, from the new United States. Not only were there the great American political documents and declarations, the Declaration of Independence and the Constitution, but also the presence of outstanding Americans in contemporary France, acting as channels through which the new ideologies were transmitted from the other side of the Atlantic. Such men as Benjamin Franklin, the first official representative of the United States in France (from 1776 to 1784)⁶⁷ and Jefferson are obvious examples. Conversely, a number of young French officers went to America in order to join military forces in the Revolutionary War against England. The best-known example is La Fayette who went equipped with a letter of recommendation from Franklin to George Washington.⁶⁸ Many other Frenchmen were able to meet the leaders of the new America personally, and to witness the consolidation of the new state on a basis of freedom.

When Boufflers had delivered his report to the Assembly, he produced a bill with the title *Loi Relative aux Découvertes utiles et aux Moyens d'en assurer la propriété aux Auteurs*, which summed up all the new ideas about inventors' rights in 18 sections.⁶⁹

The first section of the law established the new principle that each invention was the personal property of its author and that the law guaranteed the complete and unlimited right of use.⁷⁰ 'Patents of importation' could also be taken out, to cover foreign inventions introduced into France for the first time. This type of patent was indispensable in France in the century of the great mechanical discoveries, especially in view of the marked lead enjoyed by England in engineering and transport. Colbert, Turgot

⁶⁸ P. 28 of the original French report.

⁶⁷ Benjamin Franklin was an inventor of a number of practical improvements such as stoves and boats, but did not choose to take out patents, cf. *The Writings of Benjamin Franklin*, Albert H. Smith ed. (New York, 1905–1907), I, 370; IX, 183, 372 ff., 413, 445, 483, 528, 538, 569, 584, 667.

⁶⁸ *The Letters of La Fayette to Washington 1777–1799*, ed. by L. Gottschalk (New York, 1944).

⁶⁹ Cf. above, note 62.

⁷⁰ The French law did not yet make any difference between discovery (*découverte*) and invention (*invention*), of which categories, according to modern patent law, the former is not patentable.

and Boufflers had each agreed on the need for such patents, though at entirely different times. The new bill required that the inventor should deposit an exact description of the principles, means and methods of his invention, as well as drawings and models, with the office of the *Directoire de Département*. If he wished a special reward from the new fund 'for the encouragement of industry' he could file an affidavit stating that the invention was 'd'une véritable utilité'. Patents were granted for 5, 10 or 15 years, though those granted for 15 years could not be prolonged without consent of the legislation. Once granted, patents were to be registered in all provinces of the country and the public should have access to the 'catalogue of new inventions', in which the descriptions could be studied. In case of infringement the patentee had the right to confiscate the products manufactured in violation of the law and to claim damages. The patentee had the right to start commercial enterprises anywhere in the country—the guilds and royal manufactures had just been abolished—and to transfer the right of use of his invention to others. Patents, as industrial property rights, were expressly treated in the same way as *propriété mobilière*. The concept of the contract theory in patent law makes its appearance in two special sections (15 and 16) where it is emphasized that after the validity of the patent had expired, patented inventions were to belong to the public and could be used freely by any citizen anywhere in the country; this was a principle which had not previously been in the least self-evident, even though certain royal privileges had been of limited duration. According to the new law patents could become invalid for five reasons: if the description of the patent was incomplete or was wrong; if the invention was not new ('déjà consignées et décrites dans des ouvrages imprimés et publiés');⁷¹ if the invention was not put into use within two years from the day of the grant of the patent or if the cause for non-use could not be justified; and when an inventor applied for a patent for the same invention in a foreign country. This last-mentioned reason for invalidation is entirely in opposition to modern patent law.⁷²

These, then, in outline, were the proposals which Boufflers submitted

⁷¹ Except for the arbitrary examination by the French Academy of Science in special cases no general novelty search of inventions existed in France in 1791.

⁷² The requisite conditions for international cooperation in the field of patent law were first initiated in connection with the world exhibitions. The Vienna world exhibition of 1873 provided the first stimulus in this direction.

to the Constituent Assembly. The Assembly accepted this pioneering piece of French legislation on 30 December 1790. One week later, on 7 January 1791, the law came into force by the signature of Louis XVI⁷³—just two years before the king was executed as the enemy of the people.

⁷³ Original French report, p. 29.